



SIMATS Online Programming Lab
C PROGRAMMING

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QUESTIONS 25 / 25 completed

Max

MAX

Write a C program to find the largest element in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10],n,i,max=0;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7     {
8         scanf("%d",&a[i]);
9     }
10    for(i=0;i<n;i++)
11    {
12        if(a[i]>max)
13            max=a[i];
14    }
15    printf("%d is the largest element",max);
16    return 0;
17 }
```

INPUT

5
1 2 4 6 9



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Average

AVERAGE

Write a C program to calculate the average of elements in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10],i,n,sum=0;
5     float avg;
6     scanf("%d",&n);
7     for(i=0;i<n;i++)
8     {
9         scanf("%d",&a[i]);
10        sum+=a[i];
11    }
12    avg=(float)sum/n;
13    printf("average=%.2f",avg);
14    return 0;
15 }
```

INPUT

5
10 20 30 40 50



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QUESTIONS 25 / 25 completed

Reverse

REVERSE

Write a C program to reverse the elements of an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10],n,i;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7     {
8         scanf("%d",&a[i]);
9     }
10    for(i=n-1;i>=0;i--)
11    {
12        printf("%d",a[i]);
13    }
14    return 0;
15 }
```

INPUT

5
10 20 30 40 50



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QUESTIONS 25 / 25 completed

2nd Max

2ND MAX

Write a C program to find the second largest element in an array.

PROGRAM EDITOR

C

Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10],n,i,max,smax;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     max=smax=a[0];
9     for(i=1;i<n;i++)
10     {
11         if(a[i]>max)
12         {
13             smax=max;
14             max=a[i];
15         }
16         else if(a[i]>smax&& a[i]!=max)
17             smax=a[i];
18     }
19     printf("second largest=%d",smax);
20     return 0;
21 }
```

INPUT

5
10 20 30 40 50



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QUESTIONS 25 / 25 completed

Sort A-Z

SORT A-Z

Write a C program to sort an array in ascending order.

PROGRAM EDITOR

C

Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10],n,i,j,t;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     for(i=0;i<n-1;i++)
9         for(j=i+1;j<n;j++)
10             if(a[i]>a[j])
11             {
12                 t=a[i];
13                 a[i]=a[j];
14                 a[j]=t;
15             }
16     for(i=0;i<n;i++)
17         printf("%d",a[i]);
18     return 0;
19 }
```

INPUT

5
10 20 30 40 50



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Equal

EQUAL

Write a C program to check if two arrays are equal or not.

PROGRAM EDITOR

C

Run Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10],b[10],n,i;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     for(i=0;i<n;i++)
9         scanf("%d",&b[i]);
10     for(i=0;i<n;i++)
11         if(a[i]!=b[i])
12         {
13             printf("not equal");
14             return 0;
15         }
16     printf("equal");
17 }
```

INPUT

5
1 2 3 4 5
2 3 4 5 6



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Frequency

FREQUENCY

Write a C program to find the frequency of an element in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,j;
5     int a[]={2,6,6,2,3,1};
6     int size=sizeof(a)/sizeof(a[0]);
7     int fre[]={0,0,0,0,0,0};
8     for(i=0;i<size;i++)
9     {
10         if(fre[i]==1)
11             continue;
12         int count=1;
13         for(j=i+1;j<size;j++)
14             if(a[i]==a[j])
15             {
16                 fre[j]=1;
17                 count++;
18             }
19         printf("element %d occurs %d times\n",a[i],count);
20     }
21     return 0;
}
```

INPUT

Your INPUT goes here!



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Duplicate

DUPLICATE

Write a C program to remove duplicate elements from an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10],n,i,j,k;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     for(i=0;i<n;i++)
9         for(j=i+1;j<n;j++)
10             if(a[i]==a[j])
11             {
12                 for(k=j;k<n-1;k++)
13                     a[k]=a[k+1];
14                 n--;j--;
15             }
16     for(i=0;i<n;i++)
17         printf("%d",a[i]);
18     return 0;
19 }
```

INPUT

6
1 2 3 4 4



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Insert

INSERT

Write a C program to insert an element at a specific position in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10],n,i,p,x;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     scanf("%d",&p,&x);
9     for(i=n;i>p-1;i--)a[i]=a[i-1];
10     a[p-1]=x;
11     for(i=0;i<n;i++)
12         printf("%d",a[i]);
13 }
```

INPUT

5
10 20 30 40 50
3 25



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Delete

DELETE
Write a C program to delete an element from an array at a specific position.

PROGRAM EDITOR

C
Run Save

```

1 #include<stdio.h>
2 int main()
3 {
4     int a[10],n,i,p;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     scanf("%d",&p);
9     while(p<n)a[p-1]=a[p],p++;
10    for(i=0;i<n;i++)
11        printf("%d",a[i]);
12    return 0;
13 }

```

INPUT
5
10 20 30 40 50
3



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Merge

MERGE
Write a C program to merge two sorted arrays into a single sorted array.

PROGRAM EDITOR

C
Run Save

```

1 #include<stdio.h>
2 int main()
3 {
4     int a[10],b[10],c[20],n,m,i,j,k=0;
5     scanf("%d%d",&n,&m);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     for(i=0;i<m;i++)
9         scanf("%d",&b[i]);
10    for(i=0;j=0;i<n&&j<m)
11        if(a[i]<b[j])c[k++]=a[i++];
12    else c[k++]=b[j++];
13    while(i<n)c[k++]=a[i++];
14    while(j<m)c[k++]=b[j++];
15    for(i=0;i<k;i++)
16        printf("%d",c[i]);
17    return 0;
18 }

```

INPUT
3 3
1 3 5
2 4 6



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Min

MIN
Write a C program to find the smallest element in an array.

PROGRAM EDITOR

C
Run Save

```

1 #include<stdio.h>
2 int main()
3 {
4     int a[10],n,i,min;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     for(min=a[0],i=1;i<n;i++)
9         if(a[i]<min)min=a[i];
10    printf("%d",min);
11    return 0;
12 }

```

INPUT
5
10 1 2 0 4 3



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Max-Min

MAX-MIN

Write a C program to find the maximum and minimum element in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[5]={2,25,4,6,89};
5     int max,min,i;
6     max=min=a[0];
7     for(i=1;i<5;i++)
8     {
9         if(max<a[i])
10            max=a[i];
11         if(min>a[i])
12            min=a[i];
13     }
14     printf("max=%d,min=%d",max,min);
15     return 0;
16 }
```

INPUT

Your INPUT goes here!



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QUESTIONS 25 / 25 completed

Count E-O

COUNT E-O

Write a C program to count the number of even and odd elements in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[5]={10,21,30,45,50};
5     int i,odd=0,even=0;
6     for(i=0;i<5;i++)
7         if(a[i]%2==0)
8             even++;
9         else
10            odd++;
11     printf("count of odd=%d, even=%d",odd,even);
12     return 0;
13 }
14
```

INPUT

Your INPUT goes here!



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QUESTIONS 25 / 25 completed

Sum E-O

SUM E-O

Write a C program to find the sum of elements at even and odd positions in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[5]={1,3,6,7,9};
5     int i,sum=0;
6     for(i=0;i<5;i++)
7         sum+=a[i];
8     printf("sum of array=%d",sum);
9     return 0;
10 }
```

INPUT

Your INPUT goes here!



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QUESTIONS 25 / 25 completed

Product

PRODUCT

Write a C program to find the product of elements in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10],n,i,p=1;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     for(i=0;i<n;i++)
9         p*=a[i];
10    printf("%d",p);
11    return 0;
12 }
```

INPUT

5
10 4 1 2 1



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QUESTIONS 25 / 25 completed

Fun-Min-Max

FUN-MIN-MAX

Write a C program to find the largest and smallest elements in an array using functions.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2
3 int max(int a[],int n){
4     int i,m=a[0];
5     for(i=1;i<n;i++)
6         if(a[i]>m) m=a[i];
7     return m;
8 }
9
10 int min(int a[],int n){
11     int i,m=a[0];
12     for(i=1;i<n;i++)
13         if(a[i]<m) m=a[i];
14     return m;
15 }
16
17 int main(){
18     int a[10],n,i;
19     scanf("%d",&n);
20     for(i=0;i<n;i++) scanf("%d",&a[i]);
21     printf("Max=%d\nMin=%d",max(a,n),min(a,n));
22 }
```

INPUT

5
10 25 8 40 15



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QUESTIONS 25 / 25 completed

2nd Min

2ND MIN

Write a C program to find the second smallest element in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int a[10],n,i,min,smin;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)scanf("%d",&a[i]);
6     min=smin=9999;
7     for(i=0;i<n;i++)
8         if(a[i]<min){
9             smin=min;
10            min=a[i];
11        }
12     else if(a[i]<smin&&a[i]!=min)
13         smin=a[i];
14     printf("%d",smin);
15 }
```

INPUT

5
10 5 20 8 15



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QUESTIONS 25 / 25 completed

Kth Min

KTH MIN

Write a C program to find the kth smallest element in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int a[10],n,k,i,j,t;
4     scanf("%d",&n,&k);
5     for(i=0;i<n;i++)scanf("%d",&a[i]);
6
7     for(i=0;i<n;i++)
8         for(j=i+1;j<n;j++)
9             if(a[i]>a[j]){
10                 t=a[i];a[i]=a[j];a[j]=t;
11             }
12     printf("%d",a[k-1]);
13 }
```

INPUT

5 3
10 5 20 8 15



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QUESTIONS 25 / 25 completed

Missing

MISSING

Write a C program to find the missing number in an array of integers.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int a[10],n,i,s=0;
4     scanf("%d",&n);
5     for(i=0;i<n-1;i++){
6         scanf("%d",&a[i]);
7         s+=a[i];
8     }
9     printf("%d",n*(n+1)/2-s);
10 }
```

INPUT

5
1 2 3 5



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QUESTIONS 25 / 25 completed

Unique

UNIQUE

Write a C program to print all the unique elements in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int a[10],n,i,j,c;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)scanf("%d",&a[i]);
6     for(i=0;i<n;i++){
7         c=0;
8         for(j=0;j<n;j++)
9             if(a[i]==a[j])c++;
10         if(c==1)printf("%d ",a[i]);
11     }
12 }
```

INPUT

6
1 2 2 3 4 4



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QUESTIONS 25 / 25 completed
Rotate
ROTATE
Write a C program to rotate an array to the left by n positions.

C
Run Save

```

1 #include<stdio.h>
2 int main(){
3     int a[10],n,k,i,j,t;
4     scanf("%d",&n,&k);
5     for(i=0;i<n;i++)scanf("%d",&a[i]);
6     while(k--){
7         t=a[0];
8         for(i=0;i<n-1;i++)a[i]=a[i+1];
9         a[n-1]=t;
10    }
11    for(i=0;i<n;i++)printf("%d ",a[i]);
12 }

```

INPUT
5 3
1 2 3 4 5



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QUESTIONS 25 / 25 completed
Check Sort
CHECK SORT
Write a C program to check if an array is sorted in ascending or descending order.

C
Run Save

```

1 #include<stdio.h>
2 int main(){
3     int a[10],n,i,asc=1,des=1;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)scanf("%d",&a[i]);
6     for(i=0;i<n-1;i++){
7         if(a[i]>a[i+1])asc=0;
8         if(a[i]<a[i+1])des=0;
9     }
10    if(asc)printf("Ascending");
11    else if(des)printf("Descending");
12    else printf("Not Sorted");
13 }

```

INPUT
5
1 2 3 4 5



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QUESTIONS 25 / 25 completed
Pointer- Rev
POINTER- REV
Write a C program to print the elements of an array in reverse order using pointers.

C
Run Save

```

1 #include<stdio.h>
2 int main(){
3     int a[10],n,i,*p;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)scanf("%d",&a[i]);
6     p=&a[n-1];
7     for(i=0;i<n;i++)printf("%d ",*p--);
8 }

```

INPUT
5
1 20 30 40 5

QUESTIONS 25 / 25 completed

Median

MEDIAN

Write a C program to find the median of an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int a[10],n,i,j,t;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)scanf("%d",&a[i]);
6     for(i=0;i<n-1;i++)
7     for(j=i+1;j<n;j++)
8     if(a[i]>a[j]){
9         t=a[i];a[i]=a[j];a[j]=t;
10    }
11    if(n%2)
12        printf("%d",a[n/2]);
13    else
14        printf("%.1f",(a[n/2-1]+a[n/2])/2.0);
15 }
```

INPUT

```
5
9 2 5 11 9
```